Do Far-Right Mayoral Victories Affect Local Skill Composition? Evidence from a Regression Discontinuity Design in French Municipalities

MScAPEC, University of Neuchâtel

Tobias Tim Büchting

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1 Motivation and Research Question

This study investigates whether far-right victories in French municipal elections causally affect the local skill composition of the resident population. This is measured as the change in the share of tertiary-educated residents, hence residents with a bachelor's degree or higher. Estimating this relationship is challenging, because municipalities that elect far-right candidates may systematically differ from those that do not: far-right support may be correlated with income, unemployment, demographic structure, urbanization, or other unobserved factors. A simple comparison between far-right- and non-far-right-governed municipalities would therefore not necessarily identify a causal effect. A standard Regression Discontinuity Design (RDD) around the vote-share margin at which far-right lists narrowly win or lose could address selection on observables, but could still be biased if municipalities differ systematically in their outcome trends even absent a far-right victory. To address this, the study employs a Difference-in-Discontinuities (Diff-in-Disc) design, which compares the discontinuity in the change in skill composition at the far-right victory threshold, therefore removing any pre-existing discontinuity in trends unrelated to treatment. While Diff-in-Disc has previously been applied in public-finance settings, its application to far-right electoral victories and their consequences for local skill composition appears novel.

2 Contribution to the Literature

Close-election Regression Discontinuity Designs are a well-established tool for estimating the causal effects of electoral outcomes, credibly used to study legislators' policy choices, financial market reactions, and economic performance (Lee et al., 2004; Girardi, 2020; Marx et al., 2025). The validity of close-election RDD as an identification strategy has been extensively tested and confirmed across a wide range of electoral settings by Eggers et al. (2015). In addition, Hyytinen et al. (2018) use random election outcomes from electoral ties to test whether RDD can reproduce experimental results, finding that bias-corrected robust inference meets replication standards when implemented with appropriate bandwidths.

Building on this literature, this study extends this RDD and builds primarily on a more recent extension of this design: the Diff-in-Disc approach. Grembi et al. (2016) introduce this design to study Italian fiscal rules, combining a threshold discontinuity with a before/after comparison to remove any pre-existing discontinuity unrelated to treatment. Picchetti et al. (2024) formalize its identification and inference theory, showing it can reduce bias relative to a standard RDD. This thesis applies the Diff-in-Disc design to a new setting and outcome: far-right electoral victories at the local government level in France, and their effect on the change in local skill share. This combination of the novel design applied to a new electoral and socioeconomic context constitutes the core contribution of this thesis.

3 Identification Strategy and Data

This study uses a Diff-in-Disc design around the plurality-margin cutoff at which a far-right list wins a municipal election. Diff-in-Disc combines the discontinuity in outcomes at the vote-share cutoff with a comparison across pre- and post-election periods, thereby removing any pre-existing discontinuity in trends unrelated to treatment. Formally, the estimating equation takes the form:

$$Y_{it} = \alpha + \tau D_i + \beta_1 X_i + \beta_2 (D_i \times X_i) + \gamma\, Post_t + \delta\, (D_i \times Post_t) + \varepsilon_{it} \tag{1}$$

where X_i denotes the vote-share margin of the best far-right list, $D_i = 1\{X_i > 0\}$ indicates a far-right victory, and $Post_i$ indicates the post-election period. The coefficient of interest, δ, captures the effect of a far-right victory on the local skill share without any discontinuity already present at the cutoff before the election. Grembi et al. (2016) show that this design nets out confounding jumps that a standard RDD cannot separate from the treatment effect, and Picchetti et al. (2024) formalize its identification and inference properties.

This study draws on two complementary data sources. First, electoral data published by the official French Ministry of the Interior Ministère de l'Intérieur (2026), which provides open-access results for French municipal elections from 2001 onward. For each municipality, the data report vote shares for the first and, where applicable, second electoral round. Also containing list affiliations, which allow far-right lists to be identified in each contest. This study focuses on the 2008, 2014, and 2020 election waves, matched to the corresponding census periods. Second, this electoral data is merged with the socioeconomic dataset compiled by Sciences Po (2023), which provides municipality-level information on education over time. The two datasets are merged at the commune level using official INSEE municipality codes, yielding a panel data that links each municipality's electoral outcomes to its socioeconomic composition. This combined dataset allows for a precise matching of local political outcomes, such as specifically far-right electoral success, to changes in the share of tertiary-educated residents, which forms the outcome variable of interest in the empirical analysis.

4 Time Line

August	Handing in the research proposal
September	Estimate the model, get first results and finalize literature review
October	Extend the model, complete the thesis, discuss results
November	Submission first draft, review with supervisor
December	Finishing the final version
January	Defence period

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